

INFOE 4 | LAB ACTIVITY 1

Foundation of Linux Firewalls (iptables & nftables)

Step 1: Rule Inspection

Iptables (Verbose): sudo iptables -L -n -v

```
Unpacking libnftnl0:amd64 (1.0.2-3) ...
Selecting previously unselected package libnetfilter-conntrack3:amd64.
Preparing to unpack .../libnetfilter-conntrack3_1.1.0-1_amd64.deb ...
Unpacking libnetfilter-conntrack3:amd64 (1.1.0-1) ...
Selecting previously unselected package iptables.
Preparing to unpack .../iptables_1.8.11-2_amd64.deb ...
Unpacking iptables (1.8.11-2) ...
Setting up libip4tc2:amd64 (1.8.11-2) ...
Setting up libip6tc2:amd64 (1.8.11-2) ...
Setting up libnftnl0:amd64 (1.0.2-3) ...
Setting up libnetfilter-conntrack3:amd64 (1.1.0-1) ...
Setting up iptables (1.8.11-2) ...
update-alternatives: using /usr/sbin/iptables-legacy to provide /usr/sbin/iptables (iptables) in auto mode
update-alternatives: using /usr/sbin/iptables-legacy to provide /usr/sbin/iptables (iptables) in auto mode
update-alternatives: using /usr/sbin/iptables-nft to provide /usr/sbin/iptables (iptables) in auto mode
update-alternatives: using /usr/sbin/iptables-nft to provide /usr/sbin/iptables (iptables) in auto mode
update-alternatives: using /usr/sbin/arptables-nft to provide /usr/sbin/arptables (arptables) in auto mode
update-alternatives: using /usr/sbin/ebtables-nft to provide /usr/sbin/ebtables (ebtables) in auto mode
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u2) ...
root@lab1-debian:~# sudo iptables -F
root@lab1-debian:~# sudo iptables -X
root@lab1-debian:~# sudo iptables -P INPUT DROP
root@lab1-debian:~# sudo iptables -P FORWARD DROP
root@lab1-debian:~# sudo iptables -P OUTPUT ACCEPT
root@lab1-debian:~# sudo iptables -A INPUT -i lo -j ACCEPT
root@lab1-debian:~# sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
root@lab1-debian:~# sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
root@lab1-debian:~# sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
root@lab1-debian:~# sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
root@lab1-debian:~# sudo nft add table inet my_filter
root@lab1-debian:~# sudo nft add chain inet my_filter input { type filter hook input priority 0 \; policy drop \; }
root@lab1-debian:~# sudo nft add rule inet my_filter input if !o accept
root@lab1-debian:~# sudo nft add rule inet my_filter input ct state established,related accept
root@lab1-debian:~# sudo iptables -L -n -v
Chain INPUT (policy DROP 3 packets, 228 bytes)
 pkts bytes target     prot opt in     out     source            destination
  0    0 ACCEPT    all  --  *      *        0.0.0.0/0          0.0.0.0/0
  3 329 ACCEPT    all  --  *      *        0.0.0.0/0          0.0.0.0/0          ctstate RELATED,ESTABLISHED
  0    0 ACCEPT    tcp  --  *      *        0.0.0.0/0          0.0.0.0/0          tcp dpt:22
  0    0 ACCEPT    tcp  --  *      *        0.0.0.0/0          0.0.0.0/0          tcp dpt:80
  0    0 ACCEPT    tcp  --  *      *        0.0.0.0/0          0.0.0.0/0          tcp dpt:443

Chain FORWARD (policy DROP 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source            destination

Chain OUTPUT (policy ACCEPT 11 packets, 816 bytes)
 pkts bytes target     prot opt in     out     source            destination
root@lab1-debian:~#
```

Nftables (Ruleset): sudo nft list ruleset

```
root@lab1-debian:~# sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
root@lab1-debian:~# sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
root@lab1-debian:~# sudo nft add table inet my_filter
root@lab1-debian:~# sudo nft add chain inet my_filter input { type filter hook input priority 0 \; policy drop \; }
root@lab1-debian:~# sudo nft add rule inet my_filter input if !o accept
root@lab1-debian:~# sudo nft add rule inet my_filter input ct state established,related accept
root@lab1-debian:~# sudo nft add rule inet my_filter input tcp dport { 22, 80, 443 } accept
root@lab1-debian:~# sudo iptables -L -n -v
Chain INPUT (policy DROP 3 packets, 228 bytes)
 pkts bytes target     prot opt in     out     source            destination
  0    0 ACCEPT    all  --  *      *        0.0.0.0/0          0.0.0.0/0
  3 329 ACCEPT    all  --  *      *        0.0.0.0/0          0.0.0.0/0          ctstate RELATED,ESTABLISHED
  0    0 ACCEPT    tcp  --  *      *        0.0.0.0/0          0.0.0.0/0          tcp dpt:22
  0    0 ACCEPT    tcp  --  *      *        0.0.0.0/0          0.0.0.0/0          tcp dpt:80
  0    0 ACCEPT    tcp  --  *      *        0.0.0.0/0          0.0.0.0/0          tcp dpt:443

Chain FORWARD (policy DROP 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source            destination

Chain OUTPUT (policy ACCEPT 11 packets, 816 bytes)
 pkts bytes target     prot opt in     out     source            destination
root@lab1-debian:~# sudo nft list ruleset
# Warning: table ip filter is managed by iptables-nft, do not touch!
table ip filter {
  chain INPUT {
    type filter hook input priority filter; policy drop;
    ifname "lo" counter packets 0 bytes 0 accept
    ct state related,established counter packets 3 bytes 329 accept
    tcp dport 22 counter packets 0 bytes 0 accept
    tcp dport 80 counter packets 0 bytes 0 accept
    tcp dport 443 counter packets 0 bytes 0 accept
  }

  chain FORWARD {
    type filter hook forward priority filter; policy drop;
  }

  chain OUTPUT {
    type filter hook output priority filter; policy accept;
  }
}

table inet my_filter {
  chain input {
    type filter hook input priority filter; policy drop;
    if "lo" accept
    ct state established,related accept
    tcp dport { 22, 80, 443 } accept
  }
}
root@lab1-debian:~#
```

Step 2: Critical Thinking Questions

1. Stateful vs. Stateless: Why did we need the ESTABLISHED,RELATED rule? What would happen to your `sudo apt update` command if that rule were missing, even if OUTPUT is set to ACCEPT?
 - In a "Default Deny" security posture, the firewall is configured to drop all incoming traffic by default, which creates a challenge for outgoing requests like `sudo apt update`. While the server can send a request out, it will effectively be "deaf" to the response because the firewall does not recognize the incoming data as part of an existing conversation. The ESTABLISHED, RELATED rule enables stateful inspection, which allows the firewall to remember your outgoing requests and automatically permit the returning packets that belong to those specific sessions. Without this rule, your server would be unable to receive responses for any connection it initiates, causing updates and downloads to time out and fail.
2. Order of Operations: If you placed a DROP ALL rule at the very top of your INPUT chain, would your SSH rule at the bottom still work? Explain why.
 - Firewalls like iptables operate on a "First Match Wins" logic, meaning they process rules sequentially from the top of the chain to the bottom. If you were to place a DROP ALL rule at the very top of your INPUT chain, every incoming packet would match that rule immediately and be discarded. Consequently, the firewall would stop searching and never reach the SSH rule at the bottom of the list, resulting in you being locked out of your own server
3. nftables Advantage: Identify one syntax difference between iptables and nftables that makes the latter easier to manage for large-scale networks.
 - The primary advantage of nftables over the legacy iptables framework is its cleaner syntax and superior management efficiency. While iptables requires you to write separate, repetitive rules for every single port you wish to open, nftables allows for the use of braced sets. This feature enables you to combine multiple services—such as SSH, HTTP, and HTTPS—into a single rule like `tcp dport { 22, 80, 443 } accept`, which significantly simplifies the configuration for large-scale networks.

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